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Labsheet-2

1. Write a Java program to find the area of a triangle given its three sides using Heron's formula.

```
import java.util.Scanner;

public class TriangleArea {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      // Input three sides
      System.out.print("Enter side a: ");
      double a = sc.nextDouble();

      System.out.print("Enter side b: ");
      double b = sc.nextDouble();

      System.out.print("Enter side c: ");
      double c = sc.nextDouble();

      // Calculate semi-perimeter
      double s = (a + b + c) / 2;

      // Heron's formula
      double area = Math.sqrt(s * (s - a) * (s - b) * (s - c));

      // Display area
```

```
        System.out.println("Area of the triangle = " + area);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter side a: 5
Enter side b: 6
Enter side c: 7
Area of the triangle = 14.696938456699069
```

2. Create a Java program to calculate the compound interest given principal, rate, and time.

```
import java.util.Scanner;

public class CompoundInterest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input principal, rate and time
        System.out.print("Enter Principal amount: ");
        double p = sc.nextDouble();

        System.out.print("Enter Rate of interest: ");
        double r = sc.nextDouble();

        System.out.print("Enter Time (in years): ");
        double t = sc.nextDouble();
```

```
        // Calculate amount
        double amount = p * Math.pow((1 + r / 100), t);

        // Calculate compound interest
        double ci = amount - p;

        System.out.println("Compound Interest = " + ci);
        System.out.println("Total Amount = " + amount);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter Principal amount: 10000
Enter Rate of interest: 5
Enter Time (in years): 2
Compound Interest = 1025.0
Total Amount = 11025.0
```

3. Develop a Java program to find the distance between two points in a 2D plane using the distance formula.

```
import java.util.Scanner;

public class DistanceBetweenPoints {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        // Input coordinates of first point
        System.out.print("Enter x1: ");
        double x1 = sc.nextDouble();

        System.out.print("Enter y1: ");
        double y1 = sc.nextDouble();

        // Input coordinates of second point
        System.out.print("Enter x2: ");
        double x2 = sc.nextDouble();

        System.out.print("Enter y2: ");
        double y2 = sc.nextDouble();

        // Calculate distance
        double distance = Math.sqrt(
            Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2)
        );

        System.out.println("Distance between the two points = " + distance);

        sc.close();
    }
}
```

OUTPUT:-

Enter x1: 2

Enter y1: 3

Enter x2: 5

Enter y2: 7

Distance between the two points = 5.0

4. Write a Java program to demonstrate prefix and postfix increment by simulating a counter for visitors entering and leaving a store.

```
public class VisitorCounter {
   public static void main(String[] args) {

      int visitors = 0;

      // Postfix increment: first use the value, then increment
      System.out.println("Visitors before entry: " + visitors);
      System.out.println("Visitor entering (postfix): " + visitors++);
      System.out.println("Visitors after entry: " + visitors);

      // Prefix increment: first increment, then use the value
      System.out.println("\nVisitor entering (prefix): " + ++visitors);
      System.out.println("Total visitors: " + visitors);

      // Simulating a visitor leaving
      System.out.println("\nVisitor leaving...");
      visitors--;

      System.out.println("Current visitors in store: " + visitors);
```

```
    }
}
```

OUTPUT:-

Visitors before entry: 0

Visitor entering (postfix): 0

Visitors after entry: 1

Visitor entering (prefix): 2

Total visitors: 2

Visitor leaving...

Current visitors in store: 1

5. Create a Java program that takes an integer and outputs its negative value using unary operators.

```
import java.util.Scanner;

public class NegativeValue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int num = sc.nextInt();

        // Unary minus operator
        int negative = -num;
```

```
        System.out.println("Negative value = " + negative);

        sc.close();
    }
}
```

OUTPUT:-

Enter an integer: 25

Negative value = -25

6. Write a Java program to gradually reduce a number by half using /= until it becomes less than 1, counting the number of steps.

```
import java.util.Scanner;

public class HalfNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        double num = sc.nextDouble();

        int steps = 0;

        // Reduce the number by half until it becomes less than 1
        while (num >= 1) {
            num /= 2;
            steps++;
        }
```

```
        System.out.println("Final value = " + num);
        System.out.println("Number of steps = " + steps);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter a number: 16
Final value = 0.5
Number of steps = 5
```

7. Create a Java program to accumulate rainfall data for 7 days using += and find the total.

```
import java.util.Scanner;

public class Rainfall {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        double totalRainfall = 0;

        // Enter rainfall for 7 days
        for (int day = 1; day <= 7; day++) {
            System.out.print("Enter rainfall for day " + day + " (in mm): ");
            double rainfall = sc.nextDouble();

            totalRainfall += rainfall;
```

```
        }

        System.out.println("Total rainfall for 7 days = " + totalRainfall + " mm");

        sc.close();
    }
}
```

OUTPUT:-

```
Enter rainfall for day 1 (in mm): 10
Enter rainfall for day 2 (in mm): 5
Enter rainfall for day 3 (in mm): 8
Enter rainfall for day 4 (in mm): 12
Enter rainfall for day 5 (in mm): 6
Enter rainfall for day 6 (in mm): 9
Enter rainfall for day 7 (in mm): 10

Total rainfall for 7 days = 60.0 mm
```

8. Write a Java program to check whether three given angles form a valid triangle.

```
import java.util.Scanner;

public class ValidTriangle {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first angle: ");
        int angle1 = sc.nextInt();
```

```
        System.out.print("Enter second angle: ");
        int angle2 = sc.nextInt();

        System.out.print("Enter third angle: ");
        int angle3 = sc.nextInt();

        // Check validity of triangle
        if (angle1 > 0 && angle2 > 0 && angle3 > 0
              && angle1 + angle2 + angle3 == 180) {
           System.out.println("The angles form a valid triangle.");
        } else {
           System.out.println("The angles do not form a valid triangle.");
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter first angle: 60
Enter second angle: 60
Enter third angle: 60
The angles form a valid triangle.
```

9. Create a Java program to compare two strings for lexicographic order without using built-in compare functions (use relational operators on characters).

```
import java.util.Scanner;
```

```
public class LexicographicCompare {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter first string: ");
      String str1 = sc.nextLine();

      System.out.print("Enter second string: ");
      String str2 = sc.nextLine();

      int minLength = Math.min(str1.length(), str2.length());
      int result = 0;

      // Compare characters one by one
      for (int i = 0; i < minLength; i++) {
         char ch1 = str1.charAt(i);
         char ch2 = str2.charAt(i);

         if (ch1 < ch2) {
            result = -1;
            break;
         } else if (ch1 > ch2) {
            result = 1;
            break;
         }
```

```
        }

        // If all common characters are equal, compare lengths
        if (result == 0) {
            if (str1.length() < str2.length()) {
                result = -1;
            } else if (str1.length() > str2.length()) {
                result = 1;
            }
        }

        if (result == 0) {
            System.out.println("Both strings are equal.");
        } else if (result < 0) {
            System.out.println("First string comes before second string.");
        } else {
            System.out.println("First string comes after second string.");
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter first string: Apple
Enter second string: Banana
First string comes before second string.
```

10. Write a Java program to determine if a given year is both a leap year and within a given range.

```
import java.util.Scanner;

public class LeapYearRange {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter year: ");
      int year = sc.nextInt();

      System.out.print("Enter lower limit: ");
      int lower = sc.nextInt();

      System.out.print("Enter upper limit: ");
      int upper = sc.nextInt();

      // Check leap year
      boolean leapYear = (year % 400 == 0) ||
                  (year % 4 == 0 && year % 100 != 0);

      // Check if year is within range
      boolean withinRange = (year >= lower && year <= upper);

      if (leapYear && withinRange) {
         System.out.println(year + " is a leap year and is within the given range.");
```

```
        } else {
            System.out.println(year + " does not satisfy both conditions.");
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter year: 2024

Enter lower limit: 2000

Enter upper limit: 2030

2024 is a leap year and is within the given range.

11. Create a Java program to check if a student passes a course, where they must have at least 40% in theory AND 50% in practical OR overall, 50%.

```
import java.util.Scanner;

public class CourseResult {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Theory marks (%): ");
        double theory = sc.nextDouble();

        System.out.print("Enter Practical marks (%): ");
        double practical = sc.nextDouble();
```

```
        System.out.print("Enter Overall marks (%): ");
        double overall = sc.nextDouble();

        // Pass condition:
        // Theory >= 40% AND Practical >= 50%
        // OR Overall >= 50%
        if ((theory >= 40 && practical >= 50) || overall >= 50) {
            System.out.println("Student passes the course.");
        } else {
            System.out.println("Student fails the course.");
        }

        sc.close();
    }
}
```

OUTPUT:-

```
Enter Theory marks (%): 45
Enter Practical marks (%): 55
Enter Overall marks (%): 48
Student passes the course.
```

12. Write a Java program to find the smallest of four numbers using nested ternary operators.

```
import java.util.Scanner;

public class SmallestOfFour {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter four numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();
        int d = sc.nextInt();

        int smallest = (a < b)
                ? ((a < c) ? ((a < d) ? a : d) : ((c < d) ? c : d))
                : ((b < c) ? ((b < d) ? b : d) : ((c < d) ? c : d));

        System.out.println("Smallest number = " + smallest);

        sc.close();
    }
}
```

OUTPUT:-

Enter four numbers: 25 12 30 8

Smallest number = 8

13. Create a Java program to check if a character is a vowel, consonant, digit, or special symbol using the ternary operator.

```
import java.util.Scanner;

public class CharacterCheck {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        String result = (ch >= '0' && ch <= '9') ? "Digit" :
                        ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) ?
                        ((ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U' ||
                          ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
                          ? "Vowel" : "Consonant")
                        : "Special Symbol";

        System.out.println("The character is a " + result);

        sc.close();
    }
}
```

OUTPUT:-

Enter a character: A

The character is a Vowel

14. Write a Java program to swap two integers without using a temporary variable (use XOR).

```
import java.util.Scanner;

public class SwapUsingXOR {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int a = sc.nextInt();

        System.out.print("Enter second integer: ");
        int b = sc.nextInt();

        System.out.println("Before swapping: a = " + a + ", b = " + b);

        // Swapping using XOR
        a = a ^ b;
        b = a ^ b;
        a = a ^ b;

        System.out.println("After swapping: a = " + a + ", b = " + b);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter first integer: 10
Enter second integer: 20
Before swapping: a = 10, b = 20
After swapping: a = 20, b = 10.
```

15. Create a Java program to count the number of set bits (1s) in the binary representation of a number using bitwise operators.

```
import java.util.Scanner;


public class CountSetBits {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter a number: ");
      int n = sc.nextInt();


      int count = 0;
      int num = n;


      while (num > 0) {
         count = count + (num & 1);
         num = num >> 1;
      }


      System.out.println("Number of set bits = " + count);


      sc.close();
   }
}
```

OUTPUT:-

Enter a number: 13

Number of set bits = 3

16. Write a Java program to perform fast multiplication and division of a number by powers of two using shift operators.

```
import java.util.Scanner;


public class ShiftOperations {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter a number: ");
      int num = sc.nextInt();


      System.out.print("Enter the power of 2: ");
      int power = sc.nextInt();


      // Multiplication by 2^power
      int multiplication = num << power;


      // Division by 2^power
      int division = num >> power;


      System.out.println("After multiplication by 2^" + power + " = " + multiplication);
      System.out.println("After division by 2^" + power + " = " + division);


      sc.close();
   }
```

```
}
```

OUTPUT:-

```
Enter a number: 20
Enter the power of 2: 2
After multiplication by 2^2 = 80
After division by 2^2 = 5
```

17. Create a Java program to cyclically rotate the bits of an integer to the left by 2 positions.

```
import java.util.Scanner;

public class RotateBitsLeft {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter an integer: ");
      int num = sc.nextInt();

      // Cyclic left rotation by 2 positions
      int result = (num << 2) | (num >>> 30);

      System.out.println("Original number = " + num);
      System.out.println("After cyclic left rotation by 2 bits = " + result);

      sc.close();
   }
}
```

OUTPUT:-

Enter an integer: 5

Original number = 5

After cyclic left rotation by 2 bits = 20

18. Write a Java program to print the first 20 terms of the Fibonacci sequence using a for loop.

```
public class Fibonacci {
   public static void main(String[] args) {


      int first = 0;
      int second = 1;


      System.out.println("First 20 terms of Fibonacci sequence:");


      for (int i = 1; i <= 20; i++) {
         System.out.print(first + " ");


         int next = first + second;
         first = second;
         second = next;
      }
   }
}
```

OUTPUT:-

First 20 terms of Fibonacci sequence:

0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

19. Create a Java program to print all Armstrong numbers between 1 and 1000 using a for loop.

```
public class ArmstrongNumbers {
   public static void main(String[] args) {


      System.out.println("Armstrong numbers between 1 and 1000:");


      for (int num = 1; num <= 1000; num++) {
         int original = num;
         int sum = 0;


         while (original > 0) {
            int digit = original % 10;
            sum = sum + (digit * digit * digit);
            original = original / 10;
         }


         if (sum == num) {
            System.out.print(num + " ");
         }
      }
   }
}
```

OUTPUT:-

Armstrong numbers between 1 and 1000:

1 153 370 371 407 1000

20. Write a Java program to repeatedly accept a password until the correct one is entered (use do-while).

```
import java.util.Scanner;


public class PasswordCheck {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      String correctPassword = "java123";
      String password;


      do {
         System.out.print("Enter password: ");
         password = sc.nextLine();


         if (!password.equals(correctPassword)) {
            System.out.println("Incorrect password. Try again.");
         }


      } while (!password.equals(correctPassword));


      System.out.println("Correct password! Access granted.");


      sc.close();
   }
}
```

OUTPUT:-

Enter password: abc123

Incorrect password. Try again.

Enter password: test

Incorrect password. Try again.

Enter password: java123

Correct password! Access granted.

21. Create a Java program to display all factors of a number using a do-while loop.

```
import java.util.Scanner;

public class Factors {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      int i = 1;

      System.out.println("Factors of " + num + " are:");

      do {
         if (num % i == 0) {
            System.out.print(i + " ");
         }
         i++;
```

```
    } while (i <= num);

    sc.close();
  }
}
```

OUTPUT:-

```
Enter a number: 12
Factors of 12 are:
1 2 3 4 6 12
```

22. Write a Java program to reverse the digits of an integer using a while loop.

```
import java.util.Scanner;

public class ReverseNumber {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter an integer: ");
    int num = sc.nextInt();

    int reverse = 0;

    while (num != 0) {
      int digit = num % 10;
      reverse = reverse * 10 + digit;
      num = num / 10;
    }
```

```
        System.out.println("Reversed number = " + reverse);

        sc.close();
    }
}
```

OUTPUT:-

Enter an integer: 12345

Reversed number = 54321

23. Create a Java program to determine if a number is a palindrome using a while loop.

```
import java.util.Scanner;

public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int original = num;
        int reverse = 0;

        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
```

```
        }

        if (original == reverse) {
            System.out.println(original + " is a Palindrome number.");
        } else {
            System.out.println(original + " is not a Palindrome number.");
        }

        sc.close();
    }
}
```

OUTPUT:-

Enter a number: 121

121 is a Palindrome number.

24. Write a Java program to find the maximum and minimum values in an integer array using a for-each loop.

```
import java.util.Scanner;

public class MaxMinArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] arr = new int[n];
```

```
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int max = arr[0];
        int min = arr[0];

        // for-each loop
        for (int num : arr) {
            if (num > max) {
                max = num;
            }

            if (num < min) {
                min = num;
            }
        }

        System.out.println("Maximum value = " + max);
        System.out.println("Minimum value = " + min);

        sc.close();
    }
}
```

OUTPUT:-

Enter the number of elements: 5

Enter 5 elements:

10 25 5 40 15

Maximum value = 40

Minimum value = 5

25. Create a Java program to calculate the average marks of students stored in a 2D array using a for-each loop.

```
import java.util.Scanner;


public class StudentAverage {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter number of students: ");
      int students = sc.nextInt();


      System.out.print("Enter number of subjects: ");
      int subjects = sc.nextInt();


      int[][] marks = new int[students][subjects];
      int total = 0;
      int count = 0;


      System.out.println("Enter marks:");
```

```
        for (int i = 0; i < students; i++) {
            System.out.println("Student " + (i + 1) + ":");
            for (int j = 0; j < subjects; j++) {
                marks[i][j] = sc.nextInt();
            }
        }

        // for-each loop
        for (int[] student : marks) {
            for (int mark : student) {
                total += mark;
                count++;
            }
        }

        double average = (double) total / count;

        System.out.println("Average marks = " + average);

        sc.close();
    }
}
```

OUTPUT:-

```
Enter number of students: 2
Enter number of subjects: 3
Enter marks:
```

Student 1:

80 75 90

Student 2:

70 85 80

Average marks = 80.0

26. Write a Java program that takes a day number (1–7) and displays whether it’s a weekday or weekend using switch-case.

```
import java.util.Scanner;


public class DayType {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      System.out.print("Enter day number (1-7): ");
      int day = sc.nextInt();


      switch (day) {
         case 1:
         case 2:
         case 3:
         case 4:
         case 5:
            System.out.println("Weekday");
            break;


         case 6:
```

```
        case 7:
            System.out.println("Weekend");
            break;

        default:
            System.out.println("Invalid day number");
    }

    sc.close();
  }
}
```

OUTPUT:-

Enter day number (1-7): 6

Weekend

27. Create a Java program that takes a grade (A–F) and prints its qualitative meaning (e.g., A → Excellent, B → Good, etc.) using switch-case.

```
import java.util.Scanner;

public class GradeMeaning {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter grade (A-F): ");
    char grade = sc.next().toUpperCase().charAt(0);

    switch (grade) {
```

```
case 'A':
   System.out.println("Excellent");
   break;

case 'B':
   System.out.println("Good");
   break;

case 'C':
   System.out.println("Average");
   break;

case 'D':
   System.out.println("Below Average");
   break;

case 'E':
   System.out.println("Pass");
   break;

case 'F':
   System.out.println("Fail");
   break;

default:
   System.out.println("Invalid grade");
```

```
    }

    sc.close();
  }
}
```

OUTPUT:-

Enter grade (A-F): A

Excellent

28. Write a Java program to print numbers from 1 to 50, skipping all numbers that are perfect squares using continue.

```
public class SkipPerfectSquares {
  public static void main(String[] args) {

    System.out.println("Numbers from 1 to 50 excluding perfect squares:");

    for (int i = 1; i <= 50; i++) {

      int root = (int) Math.sqrt(i);

      if (root * root == i) {
        continue;
      }

      System.out.print(i + " ");
    }
  }
```

```
}
```

OUTPUT:-

Numbers from 1 to 50 excluding perfect squares:

2 3 5 6 7 8 10 11 12 13 14 15 17 18 19 20 21 22 23 24 26 27 28 29 30 31 32 33 34 35 37 38 39 40 41 42 43 44 45 46 47 48 50

29. Create a Java program to generate random integers between 1 and 100 and stop when a number divisible by both 7 and 13 is found using break.

```
import java.util.Random;


public class RandomNumberBreak {
   public static void main(String[] args) {


      Random random = new Random();


      while (true) {
         int num = random.nextInt(100) + 1;


         System.out.println("Generated number: " + num);


         if (num % 7 == 0 && num % 13 == 0) {
            System.out.println("Number divisible by both 7 and 13: " + num);
            break;
         }
      }
   }
}
```

OUTPUT:-

Generated number: 45

Generated number: 82

Generated number: 19

Generated number: 91

Generated number: 26

Generated number: 14

Generated number: 98

Generated number: 91

Generated number: 84

Number divisible by both 7 and 13: 91

30. Write a Java program that:

 Uses shift operators to check if a number is a power of 4,

 Uses bitwise to toggle the 3rd bit of that number,

 Uses a for loop to print its multiplication table but skips multiples of 6 using continue,

 Stops if it reaches a multiple of 48 using break.

```
import java.util.Scanner;

public class BitwiseAndLoop {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.print("Enter a number: ");
      int num = sc.nextInt();

      // Check if number is a power of 4 using shift operators
      boolean isPowerOf4 = false;
```

```
for (int i = 0; i < 16; i += 2) {
   if ((1 << i) == num) {
      isPowerOf4 = true;
      break;
   }
}

if (isPowerOf4) {
   System.out.println(num + " is a power of 4.");
} else {
   System.out.println(num + " is not a power of 4.");
}

// Toggle the 3rd bit (bit position 2)
int toggled = num ^ (1 << 2);

System.out.println("After toggling 3rd bit = " + toggled);

// Multiplication table
System.out.println("Multiplication table:");

for (int i = 1; i <= 10; i++) {
   int result = num * i;

   // Skip multiples of 6
```

```
        if (result % 6 == 0) {
            continue;
        }

        // Stop at multiples of 48
        if (result % 48 == 0) {
            break;
        }

        System.out.println(num + " x " + i + " = " + result);
      }

      sc.close();
   }
}
```

OUTPUT:-

Enter a number: 4

4 is a power of 4.

After toggling 3rd bit = 0

Multiplication table:

4 x 1 = 4

4 x 2 = 8

4 x 4 = 16

4 x 5 = 20

4 x 7 = 28

4 x 8 = 32

4 x 10 = 40